

	Work Instruction	Doc. Revision	01
	AXI V810 STD Y-Axis MCG Servo Motor Removal & Replacement Procedure	Effective Date	2 Oct 2016

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☐			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	08 Dec 2015	First Release	CHIN CHEE YAN
01	2 Oct 2016	Template Changes & Content Update	ONG KEAN SHENG

	Work Instruction	Doc. Revision	01
	AXI V810 STD Y-Axis MCG Servo Motor Removal & Replacement Procedure	Effective Date	2 Oct 2016

Content

Y-Axis Servo Motor Removal.....	3
Y-Axis Servo Motor Replacement.....	7

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 STD Y-Axis MCG Servo Motor Removal & Replacement Procedure	Effective Date	2 Oct 2016

Y-Axis Servo Motor Removal

1. Turn off Air Pressure Regulator.
2. Power off the SSCA main power # Important
3. Open front and rear access door for ease of outer barrier removal.
4. Disconnect all cables (Controller bulb head, Pneumatic input valve, Interlock Switch 1 and 2) as shown at Left Outer Barrier. # **Remark, label marking the Interlock Switch 1 and 2 connector before remove.**

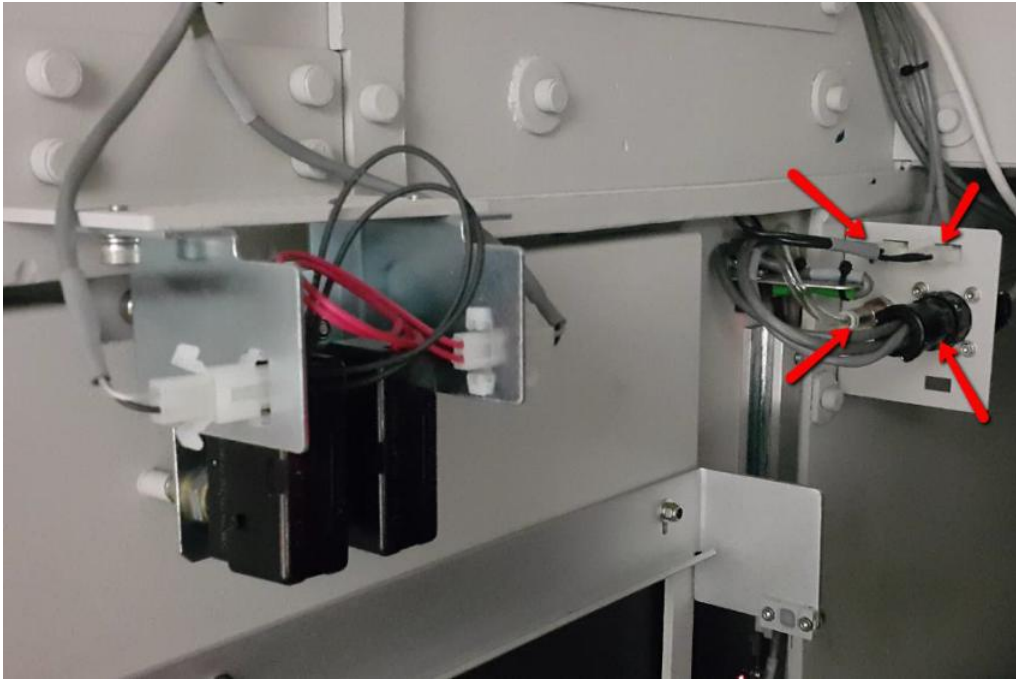


Figure 1 Bulk Head, Interlock Connector & Tubing removal

5. By using tamper proof tools, remove 4 screws & remove the Outer Barrier door.



Figure 2 Tamper Proof Screws removal

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 STD Y-Axis MCG Servo Motor Removal & Replacement Procedure	Effective Date	2 Oct 2016

6. Move the stage towards to outer barrier for ease of parts removal.
7. Move the stage near to left outer barrier and half track on Y axis ball screw for ease access and remove parts.
8. Disconnect the MCG Motor from V810 Cable, ENC/Hall, Y Axis (P/N, 4XASC-A076) and V810 Cable, Motor PWR, Y Axis (P/N, 4XASC-A077) as shown.

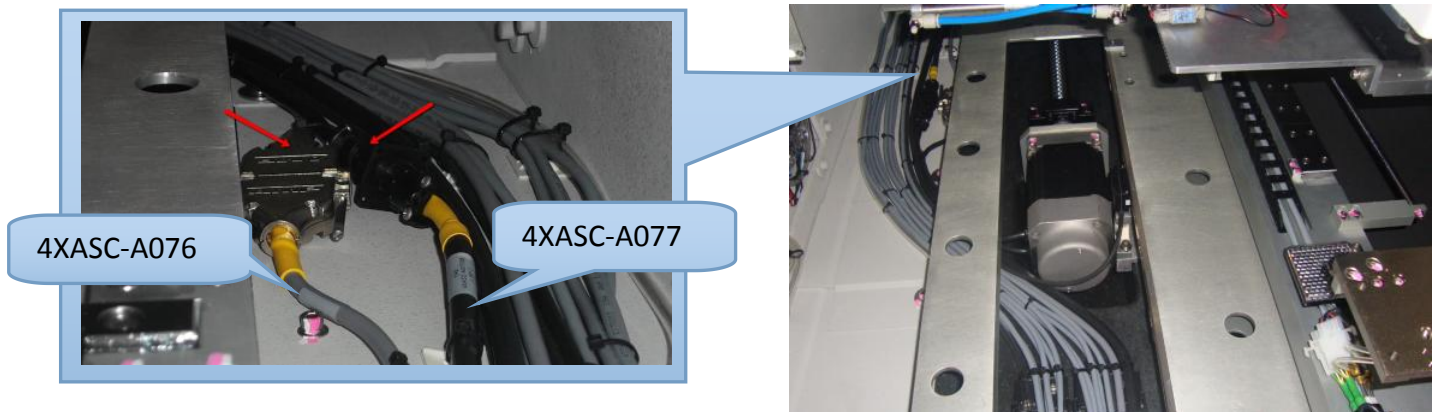


Figure 3 MCG Motor cable connector disconnect

9. Trace out teknic motor cable and remove cable tie accordingly.

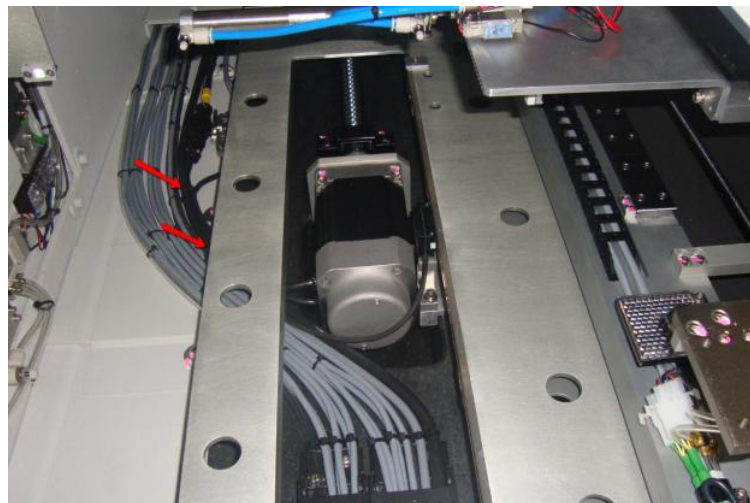


Figure 4 MCG motor cable removal

	Work Instruction	Doc. Revision	01
	AXI V810 STD Y-Axis MCG Servo Motor Removal & Replacement Procedure	Effective Date	2 Oct 2016

10. Loosen 2 x M4 screws at Bellow Coupling_Bore Dia: 12.7mm/12.7mm (P/N, 2660-0032) as shown.

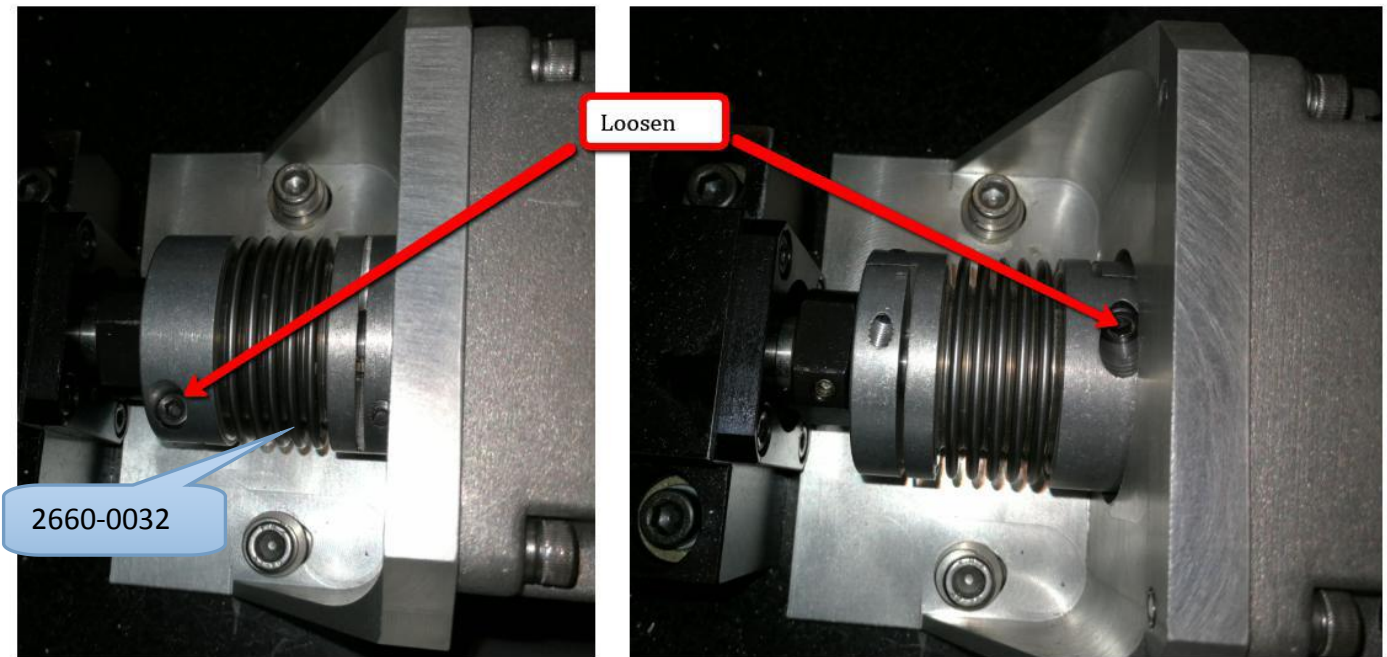


Figure 5 Coupling screw loosen

11. Unscrew (M6 x 2) as shown and gently remove the servo motor with MOTOR MOUNT, Y-AXIS (P/N, 4XAXY-011) and 2660-0032 as shown.

Remarks :

Screws require to position at visible area

Caution on handling as critical parts (Glass Scale) are within parts removal location

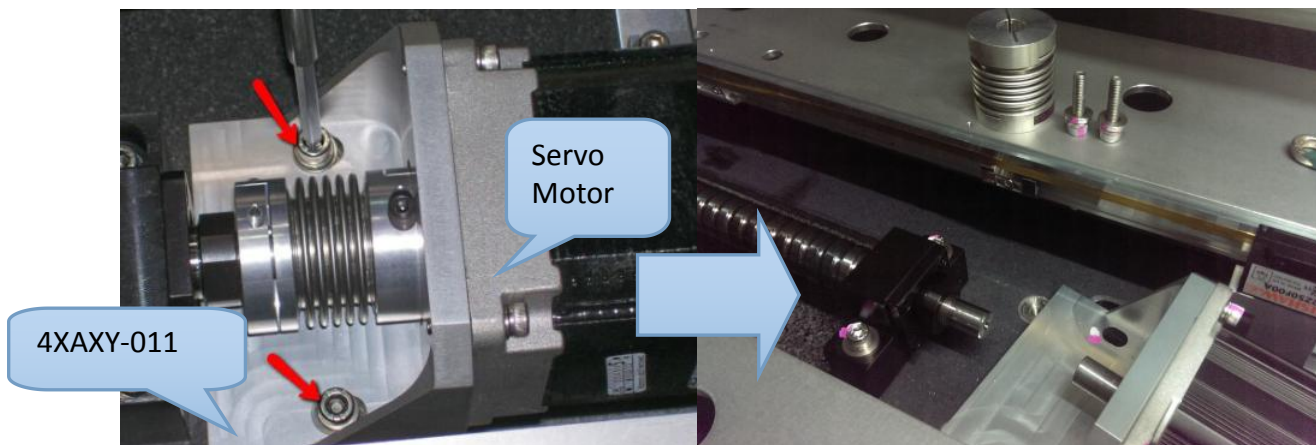


Figure 6 Servo motor removal

	Work Instruction	Doc. Revision	01
	AXI V810 STD Y-Axis MCG Servo Motor Removal & Replacement Procedure	Effective Date	2 Oct 2016

12. Gently slide out the motor and remove accordingly.

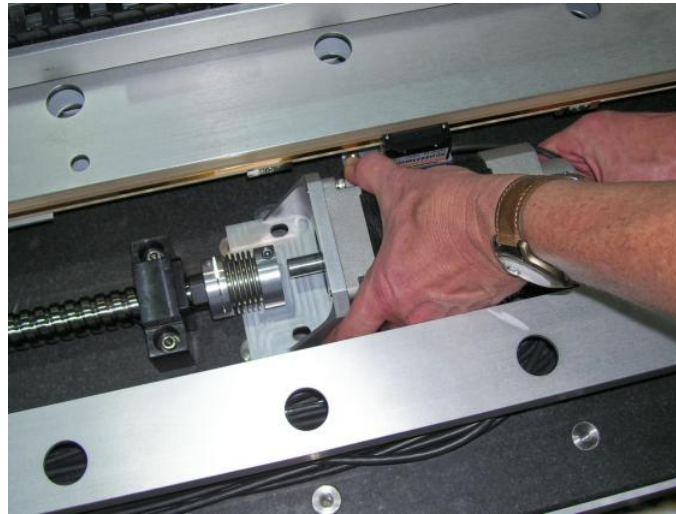


Figure 7 Servo motor removal

13. Remove the four screws (M4) that attach the motor to the MOTOR MOUNT, Y-AXIS (P/N, 4XAXY-011).

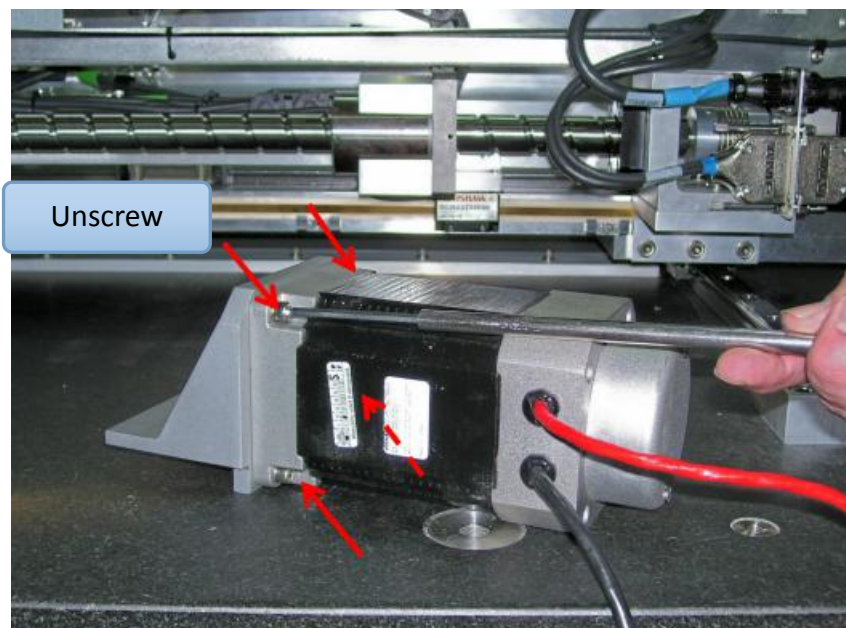


Figure 8 Motor bracket screw removal

	Work Instruction	Doc. Revision	01
	AXI V810 STD Y-Axis MCG Servo Motor Removal & Replacement Procedure	Effective Date	2 Oct 2016

Y-Axis Servo Motor Replacement

14. Prepare new V810 MOTOR, BRLSS DC SERVO ENC5000 (P/N, 4XAXY-049) as shown.



Figure 9 MCG Motor

15. Assemble the 4XAXY-049 to 4XAXY-011 and tighten 4 x screws.

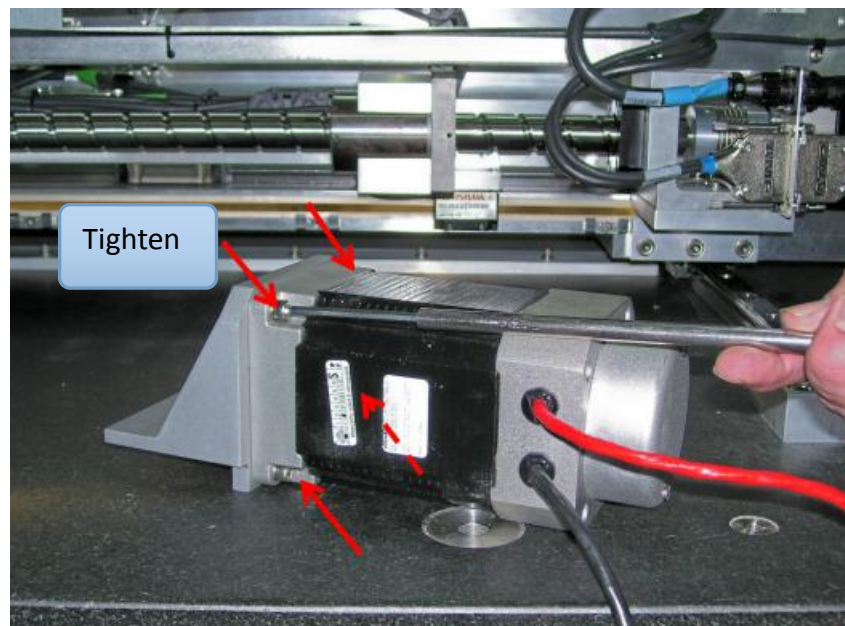


Figure 10 Bracket screw fastening

	Work Instruction	Doc. Revision	01
	AXI V810 STD Y-Axis MCG Servo Motor Removal & Replacement Procedure	Effective Date	2 Oct 2016

16. Assemble the 2660-0032 (Coupling) to V810 Y-Axis Ball screw Kit (P/N, 2650-0111) and tighten up 1 x screws as below Figure 11.

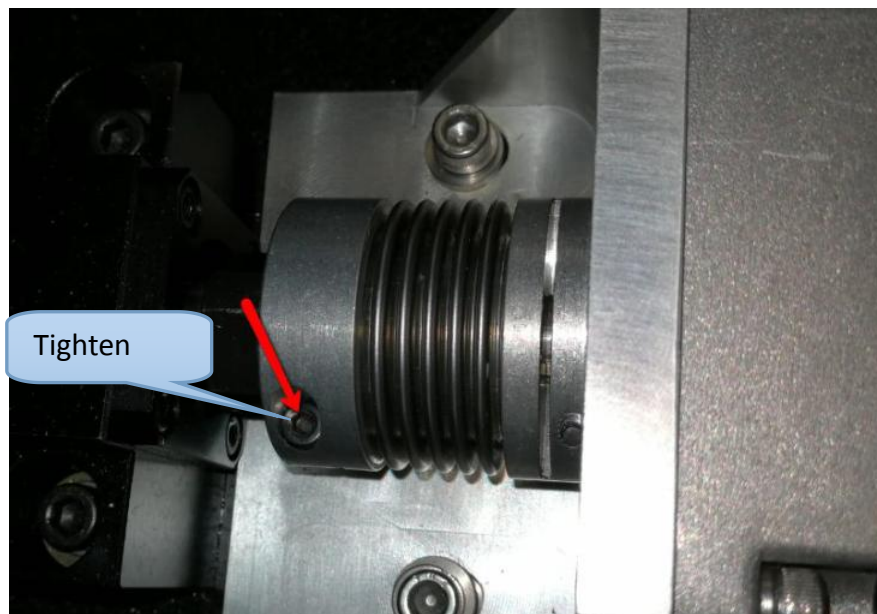


Figure 11 Coupling screw fastening

17. Support the motor weight and assemble the motor into 2660-0032 (Coupling) & tighten up 2 x screws at motor bracket as below Figure 12.

Remarks : Ensure 2 x screws at 2660-0032 (Coupling) are visible which mainly ease of access.



Figure 12 Motor bracket screw fastening

	Work Instruction	Doc. Revision	01
	AXI V810 STD Y-Axis MCG Servo Motor Removal & Replacement Procedure	Effective Date	2 Oct 2016

18. By using machinist square, Y axis Motor bracket require to align perpendicular with Y-axis base plate.

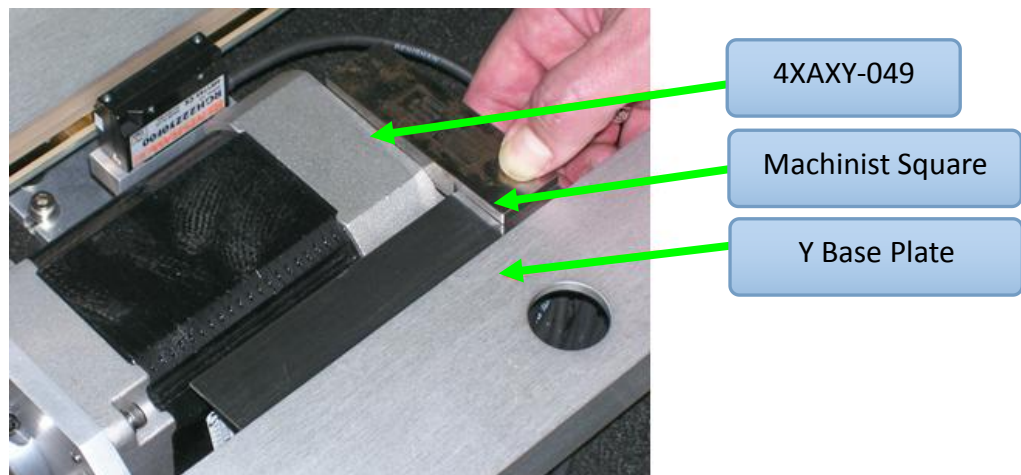


Figure 13 Perpendicular alignment

19. While holding the motor & machinist square in perpendicular position, tighten 2 x motor bracket screws. Remove the machinist square once complete.

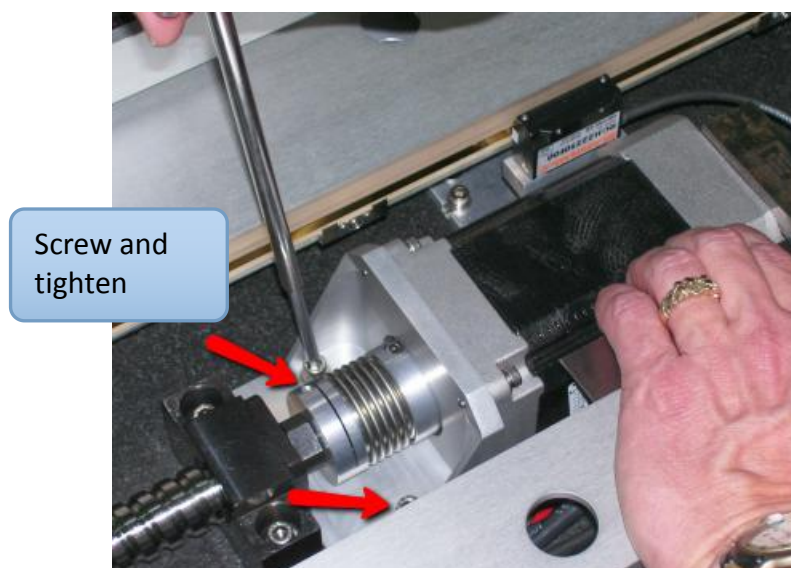


Figure 14 Motor bracket screw fastening

	Work Instruction	Doc. Revision	01
	AXI V810 STD Y-Axis MCG Servo Motor Removal & Replacement Procedure	Effective Date	2 Oct 2016

20. Tighten screw at the other end of 2660-0032. #Screw must be fully tightened.

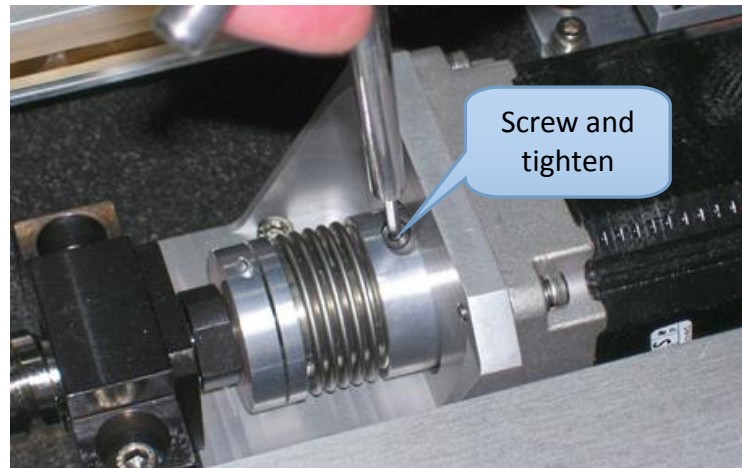


Figure 15

21. Connect the MCG motor connector to the extension cable (P/N, 4XASC-A076 and 4XASC-A077) as shown. Ensure both mating parts are completely connect

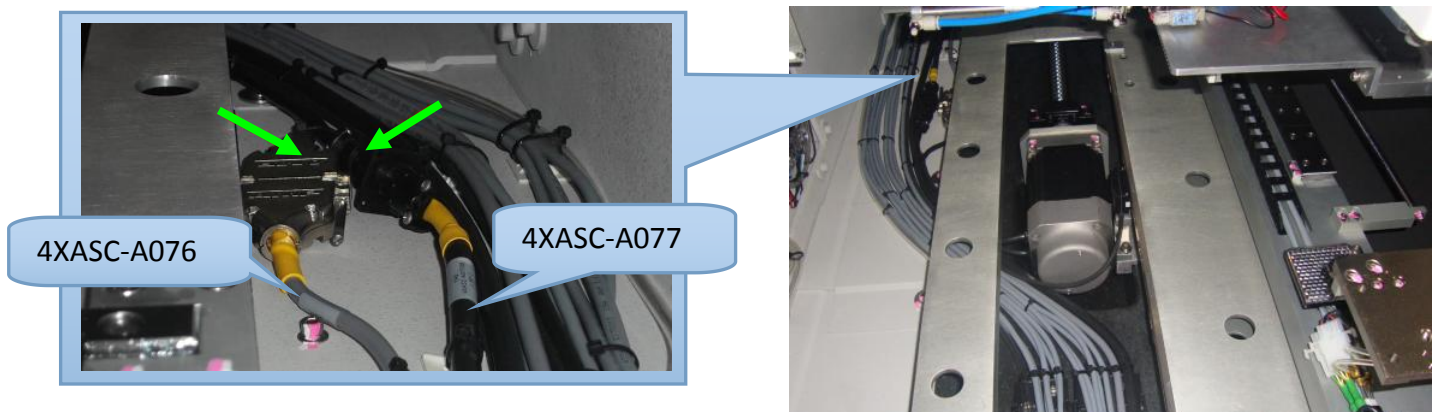


Figure 16

22. Properly route the cable and apply cable ties accordingly. Ensure cable will not have any friction stage movement. Cable tie base would be required if there is a necessary.

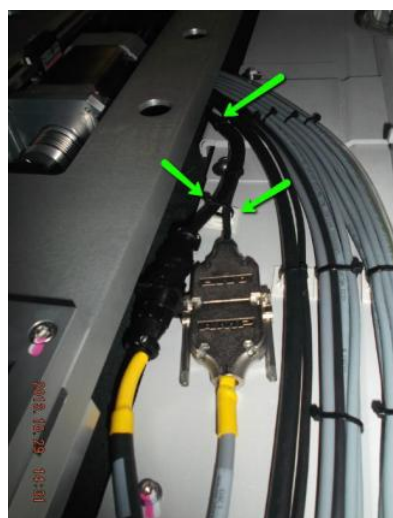


Figure 17

	Work Instruction	Doc. Revision	01
	AXI V810 STD Y-Axis MCG Servo Motor Removal & Replacement Procedure	Effective Date	2 Oct 2016

23. Remove all tools from machine stage.
24. Ensure all screws are fully tightened.
25. Manually exercise Y axis stage in “Front to Rear “or “Rear to Front” for few routes. If alignment is require, may refer to procedure **“AXI V810 STD Y-Axis Ball Screw Alignment Procedure”**
26. Reassemble the left outer barrier back to origin and reconnect all the cables connection.
27. Close all access door.
28. Go to Panel Positional, load Long scan.path and run 10 loops looping test.
29. Run 1 loop of CD & A.